

Machine–Learning Factor Investing on a Survivorship–Free US Equity Universe

Team 6 (check / replace with assigned team number)

Authors (3 columns; replace photos when laying out the PDF)

Bowen Zuo	Nicolas Couto Mota	Andrea Fontana
Person A — data & infrastructure • Walk–forward backtest engine (PIT, sector–neutral, block–gated refit) • Survivorship–free data lane (Sharadar SF1/SEP/DAILY/TICKERS) • Fama–French 5–factor + Carhart + block–bootstrap + DSR robustness	Person B — alpha model • XGBoost / Lasso / Neural Net cross–sectional regressors • Feature engineering — 14 firm features (GKX 2020 stack + chmom) • Optuna hyperparameter tuning + Diebold–Mariano model comparison	Person C — regime overlay • Hidden Markov Model regime detection (Gaussian HMM) • Gaussian Mixture Model regime detection (model comparison) • Walk–forward expanding–window OOS evaluation (look–ahead–free)

Keywords above indicate scope/contribution lanes so questions can be distributed across speakers. All three authors contributed equally in terms of effort across the 5–week project. **Note for the examiner:** Andrea Fontana is unable to attend the live presentation; the regime–overlay workstream will be summarised briefly by Bowen Zuo (engine integration owner), and the full methodology + audit script is in the appendix. Regime–internal Q&A can be taken at the high level by the present authors and a deeper follow–up arranged with Andrea after the session if needed.

Highlights

- **Full–OOS Sharpe +1.15 / FF5 α +18.7%/yr at $t=+6.85$ ($p<0.001$)** over 2012–2024 on a broad survivorship–free US equity universe (~4,400 names/month); confirming long–OOS ($Sh +0.97$, $\alpha +19.1\%$, $t=+6.00$) and test–OOS ($Sh +1.00$, $\alpha +21.2\%$, $t=+5.00$). **Honest cumulative–wealth comparison vs S&P 500 over the same window:** β –hedged pure alpha grows \$1→\$7.13 (+614%) vs S&P \$1→\$5.63 (+463%) — a 1.26× outperformance from genuine cross–sectional skill. The gross XGBoost line (\$1→\$47, +4,600%) is artificially inflated by the strategy’s +1.3 Mkt– β in a 13–year bull market and by compounding non–linearity; we lead with the β –hedged figure as the honest deployable headline.
- **Alpha survives every rigor check:** Carhart 6F (α rises to +20.1%/yr, UMD $\beta=-0.43$, momentum–averse), block–bootstrap ($P(SR\leq 0)=0.0002$), Deflated Sharpe Ratio (0.85–0.88 at $N=25$ trials), cost–grid stress (significant to ~50 bps/side), **feature–shuffle placebo** (+1.15 → −0.94 when feature→ticker mapping is permuted — rules out engine/target/cost leakage).
- **Audit–driven methodology:** caught and corrected a survivorship leak in our own engine (Sharpe +1.49 → −0.31 on S&P–500–only once PIT eligibility enforced). The honest +1.15 is what remains after rebuilding on the broad survivorship–free universe.
- **Honest down–cap finding (GKX 2020 §IV.D):** on the strict rolling top–2,000 large–cap subset, FF5 α collapses to +1.8%/yr at $t=0.96$ (n.s.) — alpha lives in the small/mid–cap tail. **Cost–realism: we beat S&P on Sharpe up to 25 bps/side** (4× our 10 bps headline). At 10–15 bps (Frazzini–Israel–Moskowitz 2018 estimate is ~6 bps/side for AQR US–equity at \$100B AUM) our Sharpe is +1.07→+1.14 vs S&P’s +0.99. Capacity binds at very large AUM; regime overlay is universe–dependent (works on strict–S&P, net–zero on broad due to COVID monthly–lag).

Brief Report

Scope. We test whether **machine–learning cross–sectional alpha** on US equities survives survivorship–bias correction, PIT eligibility filters, realistic costs, and multiple–testing penalties on a single ~13–year OOS path (2012–2024). Pipeline: monthly–rebalanced dollar–neutral long–short with **sector–neutral construction** ($k=20$ per GICS sector ≈ 440 positions), trained via **walk–forward expanding–window cross–validation** (120–month training window, block–gated refit every 12 months).

Method. Three models with a shared `fit/predict` interface — **Lasso** (regularised linear baseline), **XGBoost** (canonical gradient–boosted ensemble, **Optuna**–tuned via TPE on a 2017–18 validation window), and a small **PyTorch NN** — on a **14–feature GKX–style panel** (price–trend, volatility, liquidity, value, quality, plus `chmom`). Target = **sector–relative monthly return**. Evaluation: annualised Sharpe (Newey–West–corrected), **FF5 alpha with HAC SE**, information coefficient (Spearman). Model comparison via **Diebold–Mariano**; XGBoost wins on every window. Robustness battery: 3/3 **sanity gate** (Random/Oracle/Uniform on synthetic panel), block–bootstrap (6–mo blocks), DSR at $N=25$, **Carhart 6F momentum control**, dense **k–sweep** ($k \in [1,100]$ + bootstrap–CI plateau zoom), single–name leave–one–out, and the **feature–shuffle placebo** (the cleanest leakage test). Regime overlay: walk–forward Gaussian **HMM** on 6 macro–financial features, leverage–only rule.

What was left out: long–form audit narrative (in `PIT_INVESTIGATION_REPORT.pdf`); per–window FF3 regressions; Optuna trial–by–trial logs; the regime HMM’s IS–vs–OOS crisis–detection–rate breakdown (91.7% / 51.1%); pre–correction lineage Phases 1–15.

Main references (beyond course material on Lasso / gradient boosting / NN / cross–validation / hyperparameter tuning): Gu, Kelly & Xiu (2020) Empirical Asset Pricing via Machine Learning, RFS 33(5) — methodological template + feature set; Bailey & López de Prado (2014) — Deflated Sharpe Ratio; Fama & French (2015) — FF5 model; Carhart (1997) — momentum control; Newey & West (1987) — HAC standard errors; **Sharadar/Nasdaq Data Link** — primary data source under licence. Full citation list in the appendix.

Description of Appendices

Optional appendix material (single combined PDF, in the order it appears): full long–form report with all tables and figures (`report/REPORT.md`, ~30 pages when typeset); the project decision log with chronological provenance for every methodological choice (`DECISIONS.md`, 1,233 lines); the survivorship–leak audit and methodology corrections (`PIT_INVESTIGATION_REPORT.pdf`); robustness–check artefacts (Phase 25 Deflated Sharpe + bootstrap; Phase 26 leave–one–out; Phase 27/27b dense k–sweep with CI plateau); regime–overlay ablation and overlay–failure diagnostic; all source code under `src/` and `notebooks/{persona, personb, personc}/`. **Total submitted PDF size:** (to be measured after compilation; target < 30 pages, single PDF).